

16 *Use of the Data Booklet is relevant to this question.*

Which volume of $\text{O}_2(\text{g})$, measured at room temperature and pressure, reacts with 1.0 g of methane to produce $\text{H}_2\text{O}(\text{l})$ and equal volumes of $\text{CO}(\text{g})$ and $\text{CO}_2(\text{g})$?

- A** 1.31 dm^3 **B** 2.63 dm^3 **C** 5.25 dm^3 **D** 42.0 dm^3